Project: Tarform Motorcycle

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CHANGE HISTORY

PAGE No.	CHANGES IN SCHEMATICS	PREVIOUS PART NUMBER	NEW PART NUMBER
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Rivision History

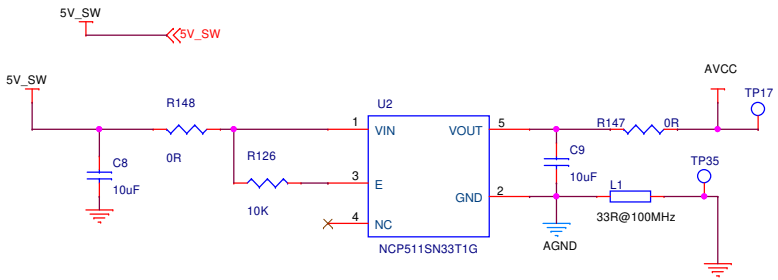
Version	DATE	Modified By	Reviewed By	Modification/comment
V1.0	11/12/2020	Vinit	Sreekala	Intial Draft
V1.1	23/02/2021	Vinit	Vinit / Narendra	Added ESD protection on SIM connector and updated ESD protection on SD card connector to 8 channel Added external antenna connector for WIFI and LTE module Updated review comments from PCB layout engineer to correct net list errors U27 IC part number updated to RClamp3304P J10 and J11 connector part number modified to DW-05-09-F-D-400

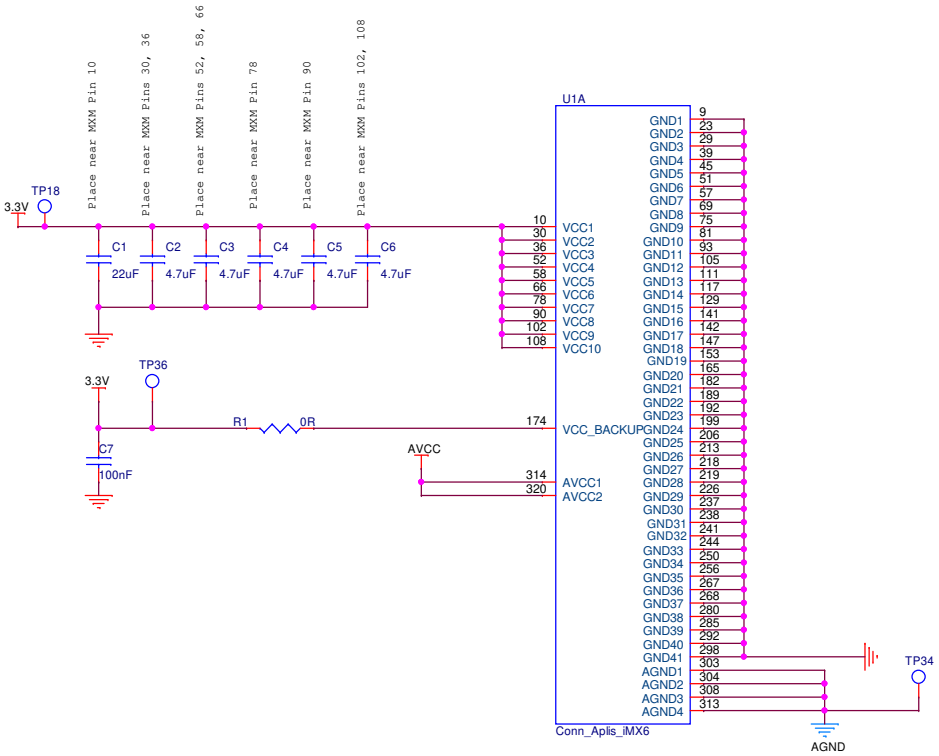
TARFORM

Title: Index

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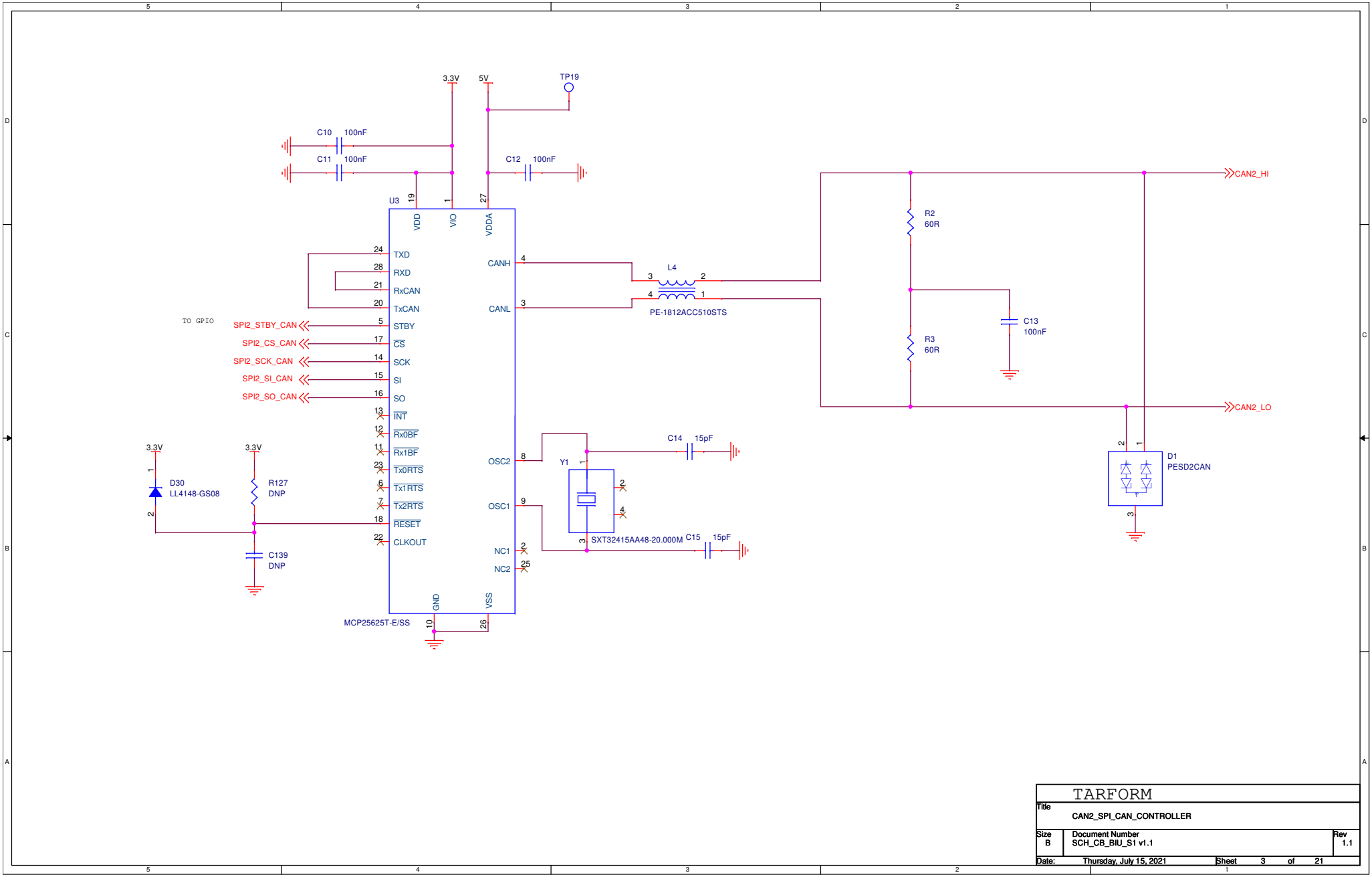
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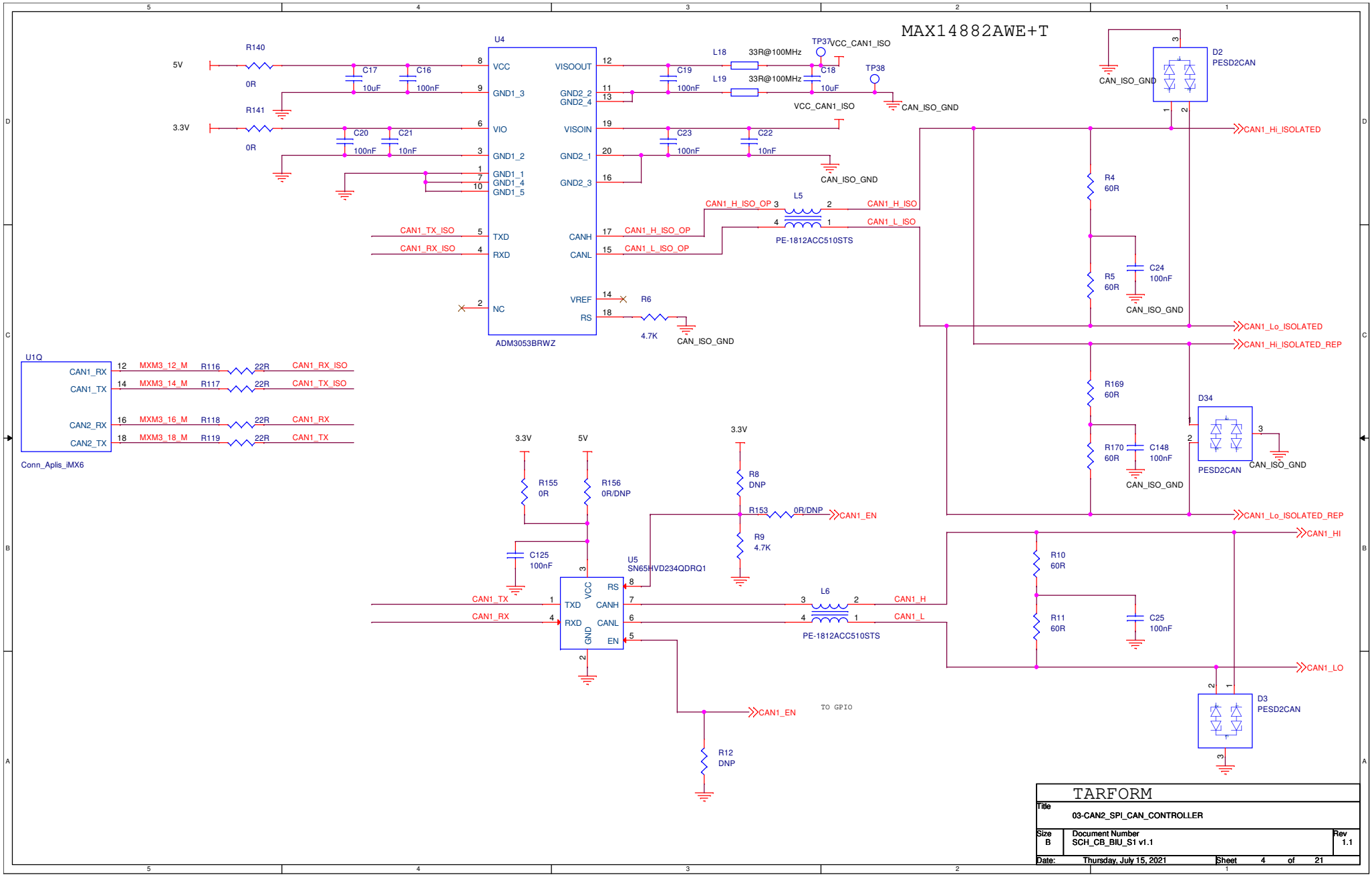


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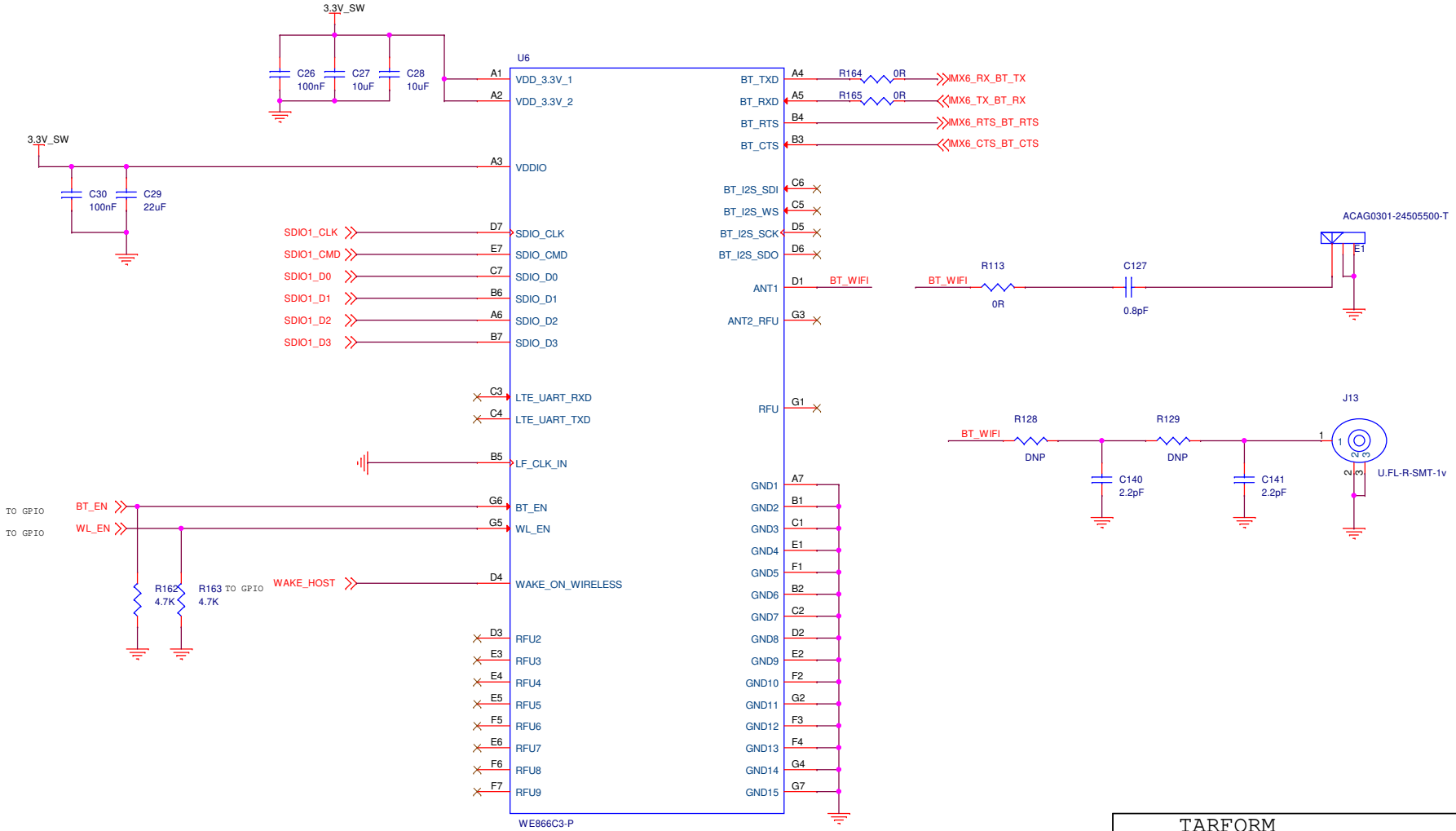
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Title	CAN2_SPI_CAN_CONTROLLER	
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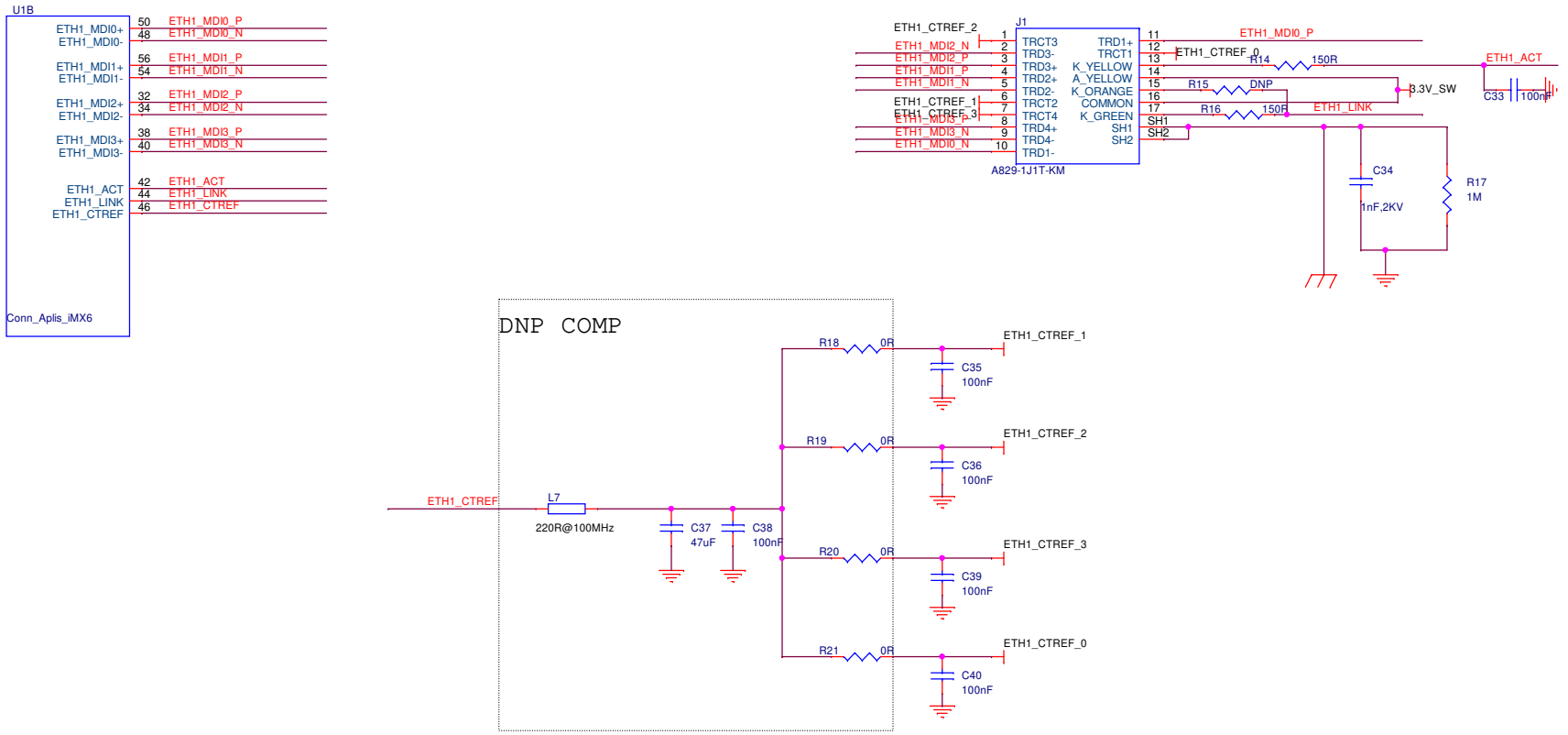

MAX14882AWE+T
U4
ADM3053BRWZ
VCC
GND1_3
VIO
GND1_2
GND1_1
GND1_4
GND1_5
NC
TXD
RXD
VISOOUT
GND2_2
GND2_4
VISOIN
GND2_1
GND2_3
CANH
CANL
VREF
RS
R140
0R
R141
0R
C17
10uF
C16
100nF
C20
100nF
C21
10nF
C19
100nF
C18
10uF
C23
100nF
C22
10nF
L18
33R@100MHz
L19
33R@100MHz
TP37
TP38
VCC_CAN1_ISO
CAN_ISO_GND
R6
4.7K
CAN1_TX_ISO
CAN1_RX_ISO
CAN1_H_ISO_OP
CAN1_L_ISO_OP
L5
PE-1812ACC510STS
CAN1_H_ISO
CAN1_L_ISO
U1Q
Conn_Aplis_iMX6
MXM3_12_M
MXM3_14_M
MXM3_16_M
MXM3_18_M
R116
R117
R118
R119
22R
CAN1_RX_ISO
CAN1_TX_ISO
CAN1_RX
CAN1_TX
U5
SN65HVD234QDRQ1
R155
0R
R156
0R/DNP
R8
DNP
R153
0R/DNP
R9
4.7K
C125
100nF
CAN1_EN
TO GPIO
R12
DNP
L6
PE-1812ACC510STS
CAN1_H
CAN1_L
D2
PESD2CAN
D3
PESD2CAN
D34
PESD2CAN
R4
60R
R5
60R
C24
100nF
R169
60R
R170
60R
C148
100nF
R10
60R
R11
60R
C25
100nF
CAN1_Hi_ISOLATED
CAN1_Lo_ISOLATED
CAN1_Hi_ISOLATED_REP
CAN1_Lo_ISOLATED_REP
CAN1_HI
CAN1_LO
TARFORM
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03-CAN2_SPI_CAN_CONTROLLER
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Bluetooth And Wi-Fi module

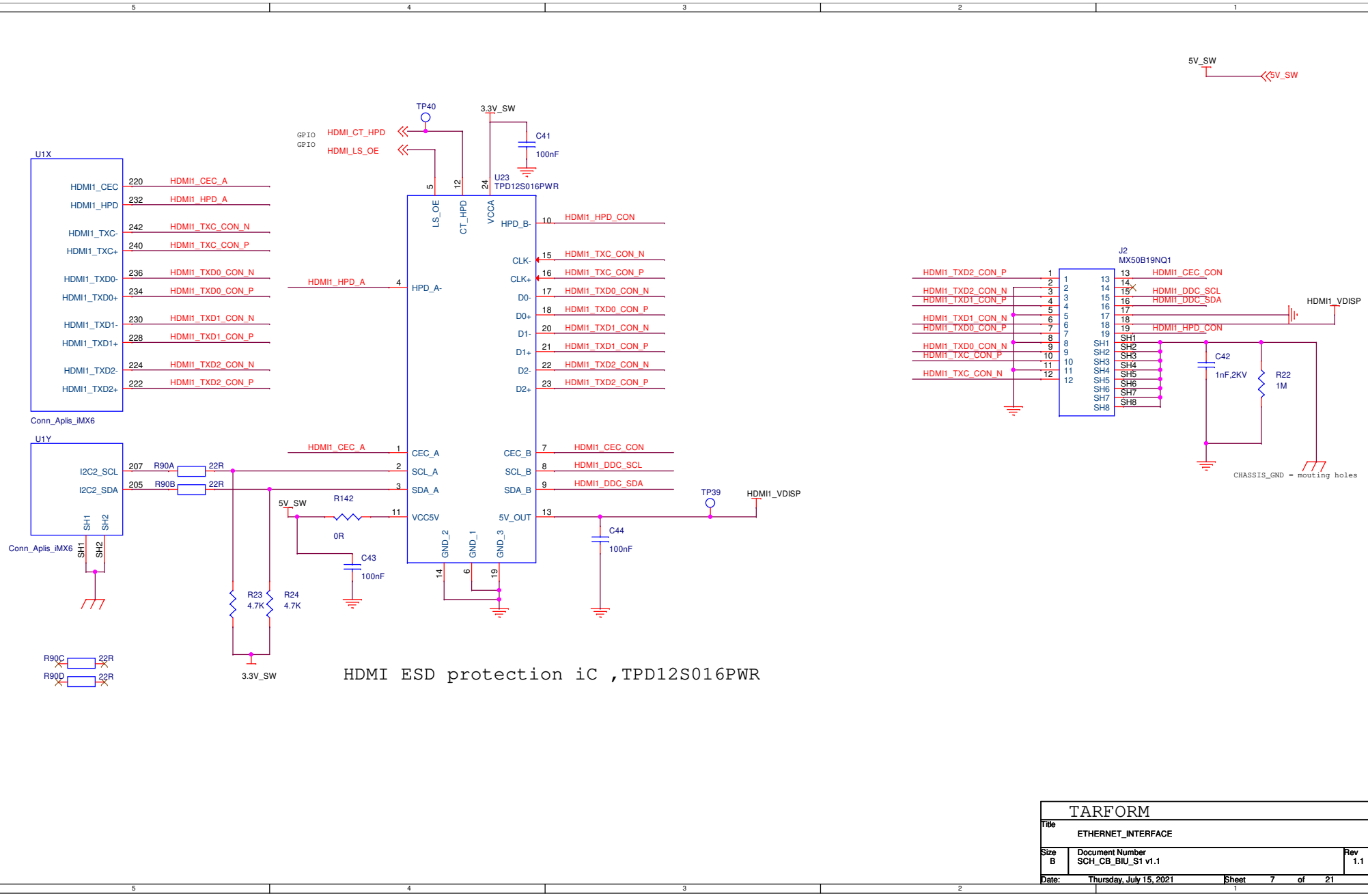


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Title: BLUETTOTH_WIFI		
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ETHERNET INTERFACE

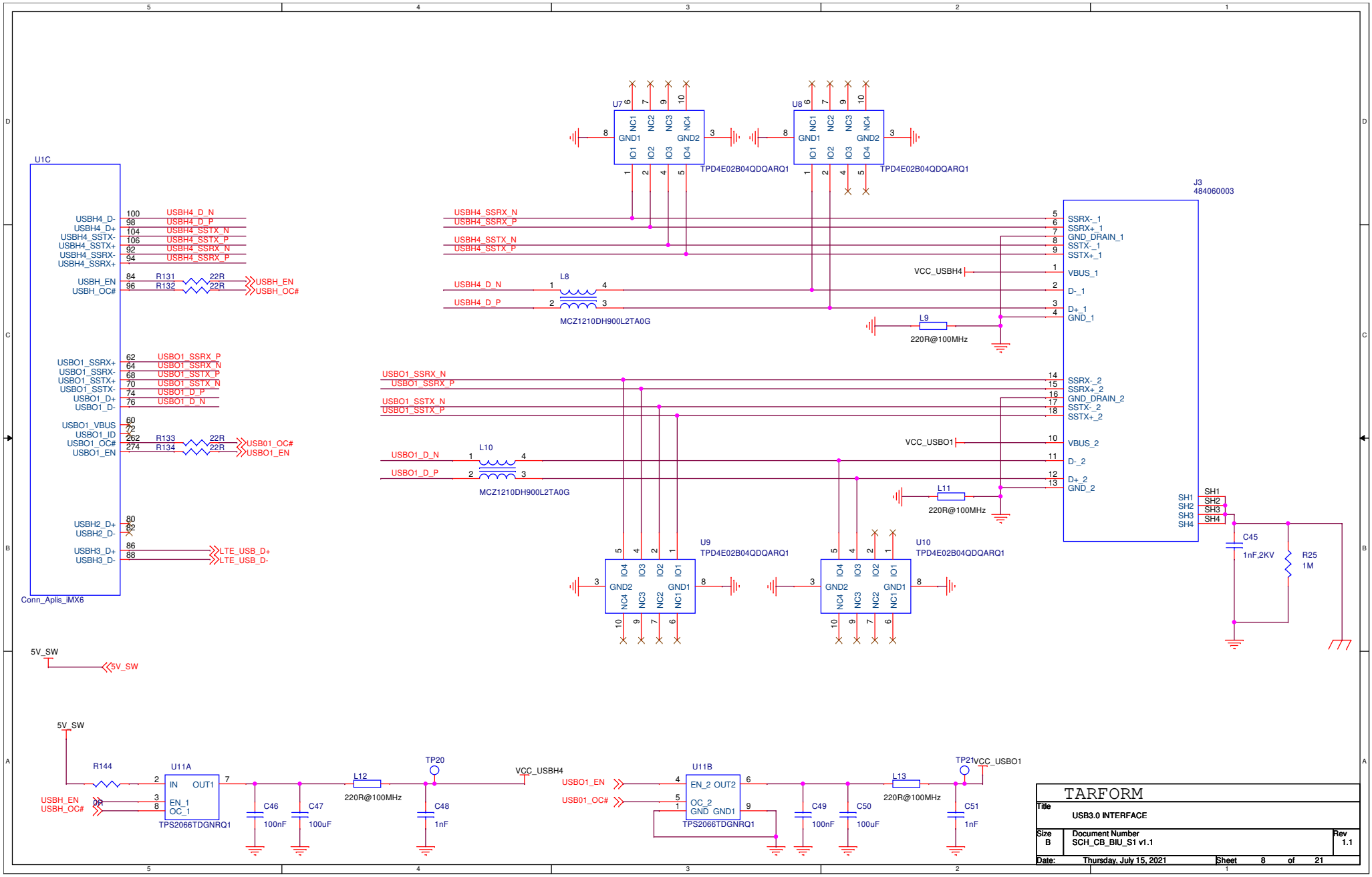


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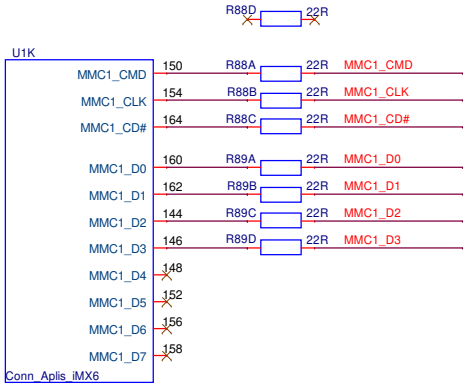


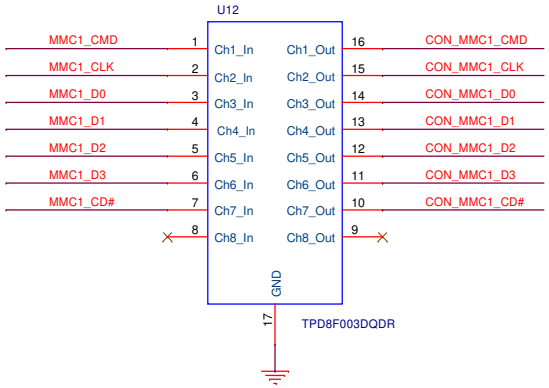
HDMI ESD protection iC ,TPD12S016PWR

TARFORM		
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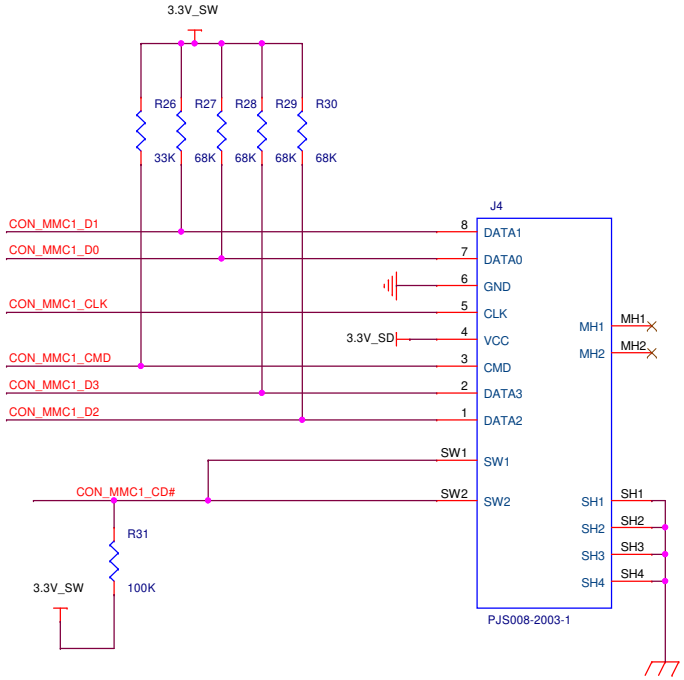

U1C
Conn_Aplis_iMX6
USBH4_D-
USBH4_D+
USBH4_SSTX-
USBH4_SSTX+
USBH4_SSRX-
USBH4_SSRX+
USBH4_D_N
USBH4_D_P
USBH4_SSTX_N
USBH4_SSTX_P
USBH4_SSRX_N
USBH4_SSRX_P
USBH_EN
USBH_OC#
R131 22R
R132 22R
USBO1_SSRX+
USBO1_SSRX-
USBO1_SSTX+
USBO1_SSTX-
USBO1_D+
USBO1_D-
USBO1_SSRX_P
USBO1_SSRX_N
USBO1_SSTX_P
USBO1_SSTX_N
USBO1_D_P
USBO1_D_N
USBO1_VBUS
USBO1_ID
USBO1_OC#
USBO1_EN
R133 22R
R134 22R
USBH2_D+
USBH2_D-
USBH3_D+
USBH3_D-
LTE_USB_D+
LTE_USB_D-
U7
U8
TPD4E02B04QDQARQ1
L8
MCZ1210DH900L2TA0G
L9
220R@100MHz
J3
484060003
SSRX-_1
SSRX+_1
GND_DRAIN_1
SSTX-_1
SSTX+_1
VBUS_1
D-_1
D+_1
GND_1
SSRX-_2
SSRX+_2
GND_DRAIN_2
SSTX-_2
SSTX+_2
VBUS_2
D-_2
D+_2
GND_2
L10
L11
U9
U10
SH1
SH2
SH3
SH4
C45
1nF,2KV
R25
1M
5V_SW
R144
U11A
TPS2066TDGNRQ1
IN
OUT1
EN_1
OC_1
C46
100nF
C47
100uF
L12
220R@100MHz
TP20
C48
1nF
VCC_USBH4
U11B
EN_2
OUT2
OC_2
GND
GND_1
C49
100nF
C50
100uF
L13
220R@100MHz
TP21
C51
1nF
VCC_USBO1
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USB3.0 INTERFACE
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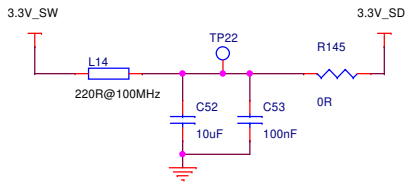
MICRO SD CARD INTERFACE





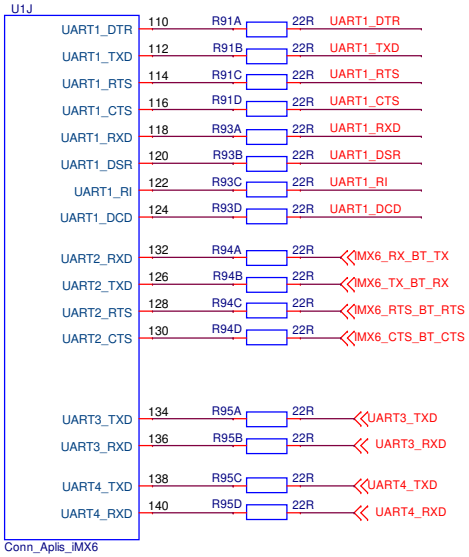
ESD protection

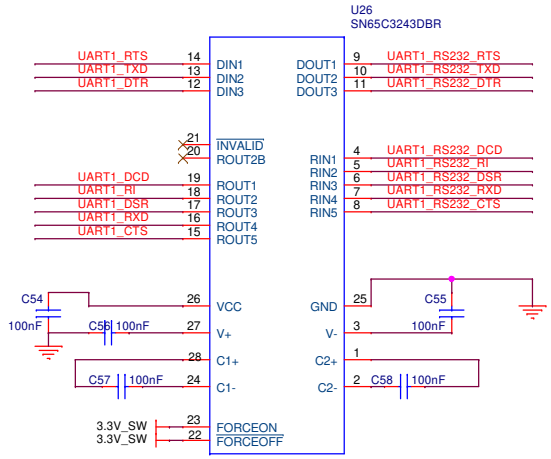


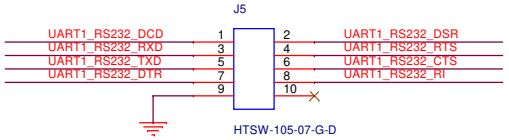


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Title: MICRO SD CARD INTERFACE		
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UART INTERFACE







CONNECTOR PENDING

UART1 – FOR DEBUG

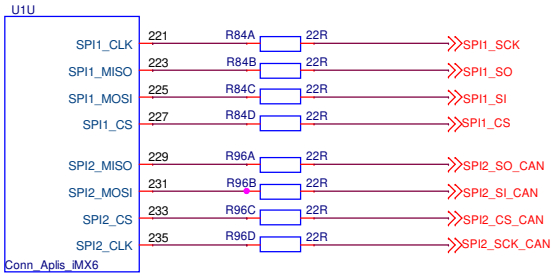
UART2 – FOR BT

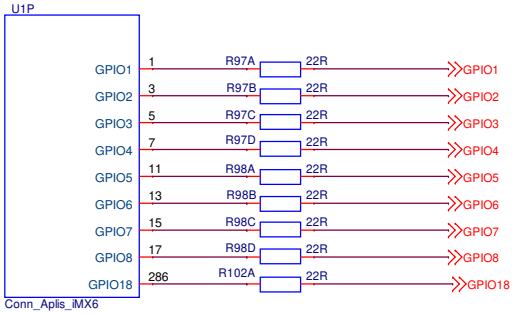
UART3 – FOR FUTURE USE

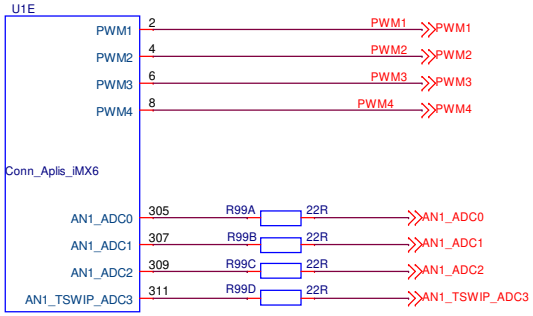
UART4 – FOR FUTURE USE

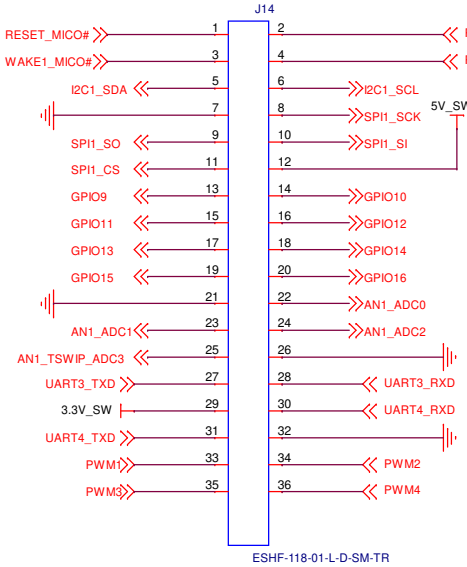
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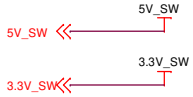
FUTURE USE CONNECTOR

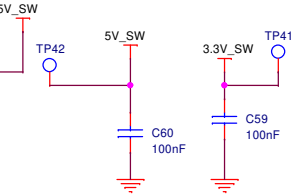






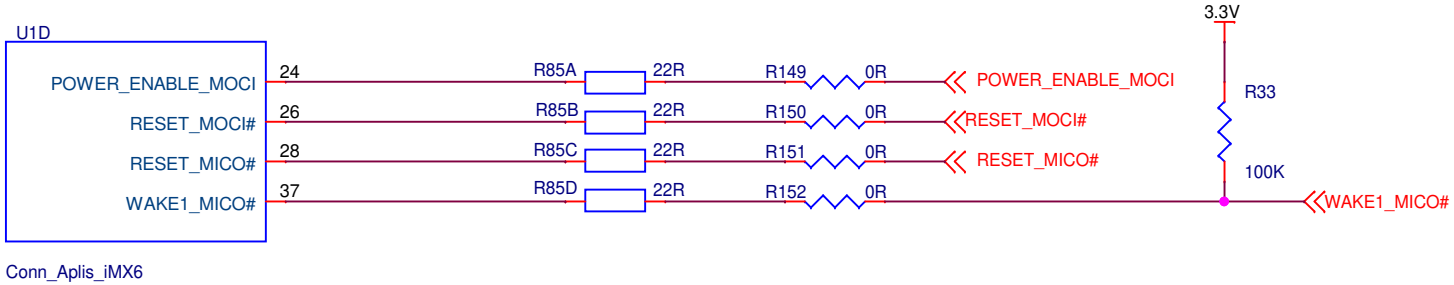


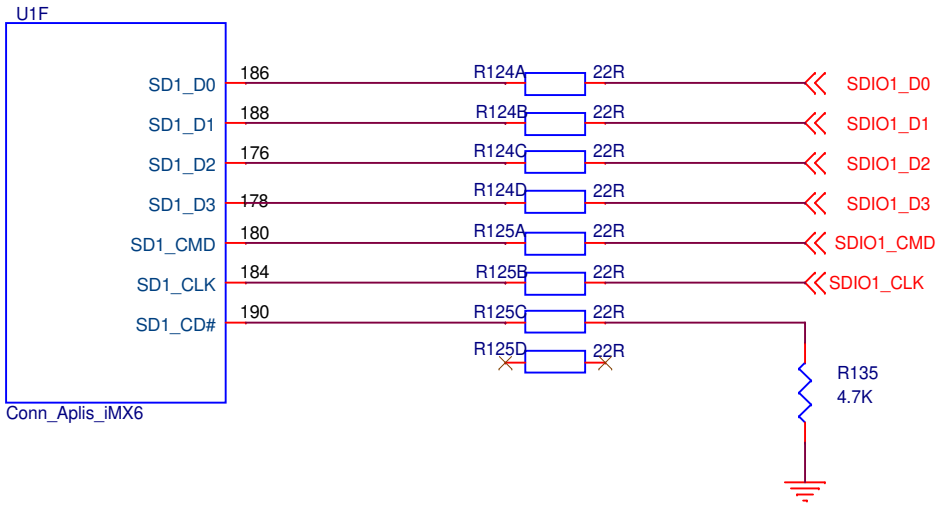




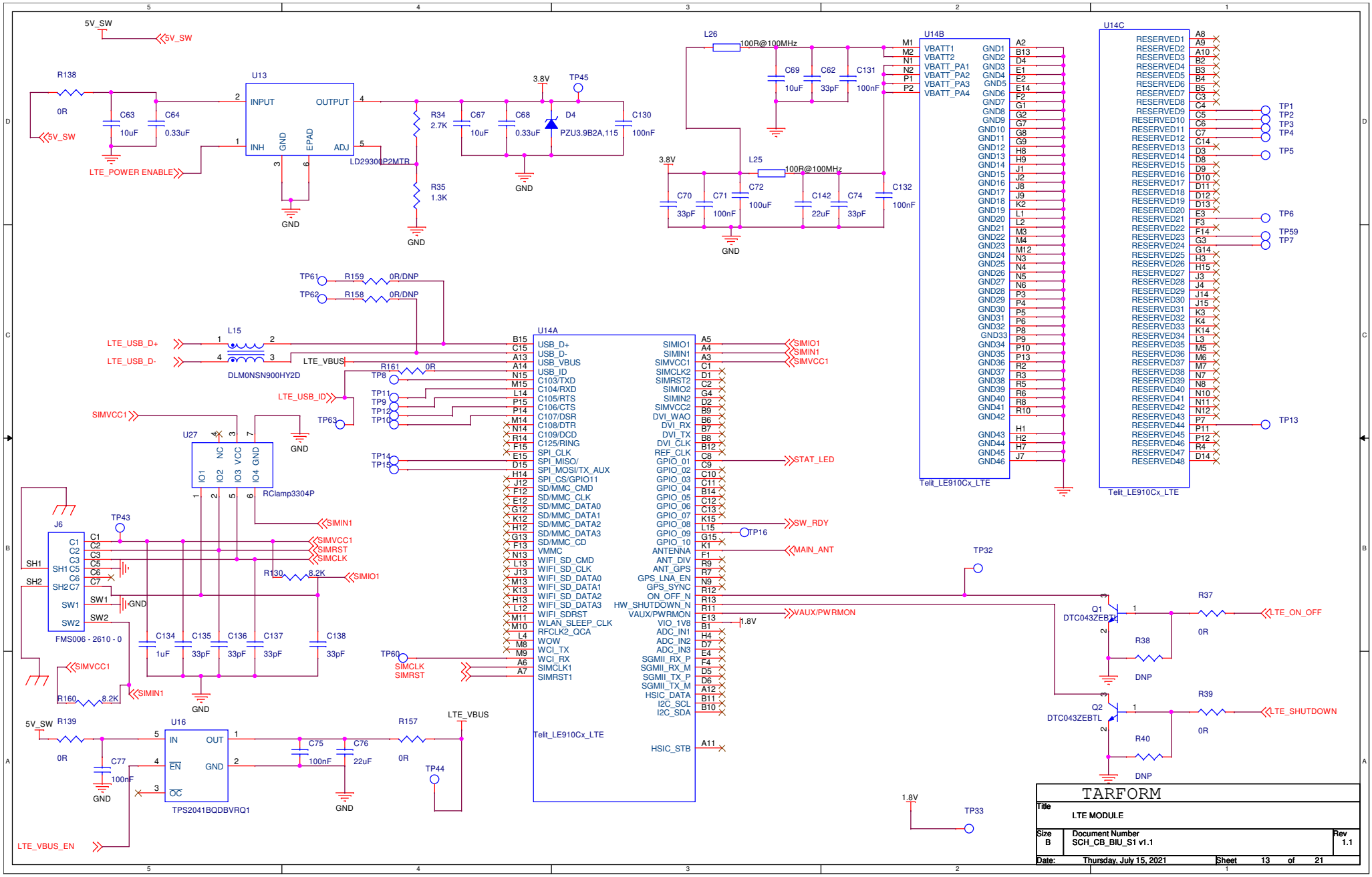
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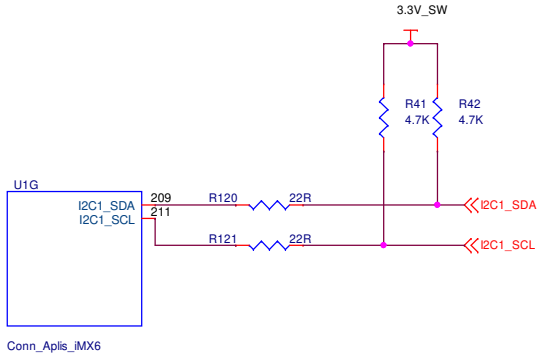
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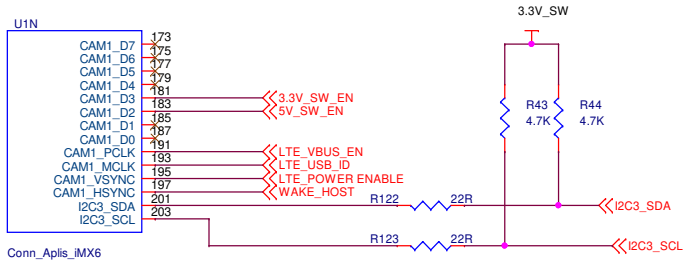


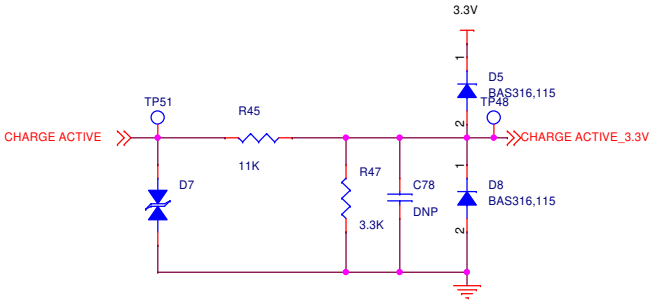


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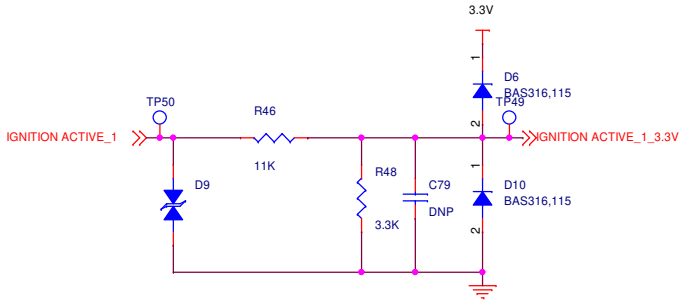

5V_SW
R138
0R
C63
10uF
C64
0.33uF
U13
INPUT
OUTPUT
INH
GND
EPAD
ADJ
LD29300P2MTR
LTE_POWER ENABLE
R34
2.7K
C67
10uF
C68
0.33uF
D4
PZU3.9B2A,115
C130
100nF
3.8V
TP45
R35
1.3K
L26
100R@100MHz
C69
10uF
C62
33pF
C131
100nF
L25
100R@100MHz
C70
33pF
C71
100nF
C72
100uF
C142
22uF
C74
33pF
C132
100nF
U14B
Telit_LE910Cx_LTE
U14C
Telit_LE910Cx_LTE
TP1
TP2
TP3
TP4
TP5
TP6
TP59
TP7
TP13
TP61
TP62
R159
0R/DNP
R158
0R/DNP
L15
DLM0NSN900HY2D
LTE_USB_D+
LTE_USB_D-
LTE_VBUS
R161
0R
TP8
LTE_USB_ID
TP11
TP9
TP12
TP10
TP63
SIMVCC1
U27
RClamp3304P
TP14
TP15
U14A
USB_D+
USB_D-
USB_VBUS
USB_ID
C103/TXD
C104/RXD
C105/RTS
C106/CTS
C107/DSR
C108/DTR
C109/DCD
C125/RING
SPI_CLK
SPI_MISO/
SPI_MOSI/TX_AUX
SPI_CS/GPIO11
SD/MMC_CMD
SD/MMC_CLK
SD/MMC_DATA0
SD/MMC_DATA1
SD/MMC_DATA2
SD/MMC_DATA3
SD/MMC_CD
VMMC
WIFI_SD_CMD
WIFI_SD_CLK
WIFI_SD_DATA0
WIFI_SD_DATA1
WIFI_SD_DATA2
WIFI_SD_DATA3
WIFI_SDRST
WLAN_SLEEP_CLK
RFCLK2_QCA
WOW
WCI_TX
WCI_RX
SIMCLK1
SIMRST1
SIMIO1
SIMIN1
SIMVCC1
SIMVCC2
DVI_WAO
DVI_RX
DVI_TX
DVI_CLK
REF_CLK
GPIO_01
GPIO_02
GPIO_03
GPIO_04
GPIO_05
GPIO_06
GPIO_07
GPIO_08
GPIO_09
GPIO_10
ANTENNA
ANT_DIV
ANT_GPS
GPS_LNA_EN
GPS_SYNC
ON_OFF_N
HW_SHUTDOWN_N
VAUX/PWRMON
VIO_1V8
ADC_IN1
ADC_IN2
ADC_IN3
SGMII_RX_P
SGMII_RX_M
SGMII_TX_P
SGMII_TX_M
HSIC_DATA
I2C_SCL
I2C_SDA
HSIC_STB
Telit_LE910Cx_LTE
STAT_LED
SW_RDY
TP16
MAIN_ANT
1.8V
TP32
VAUX/PWRMON
J6
FMS006 - 2610 - 0
SH1
SH2
SW1
SW2
GND
C134
1uF
C135
33pF
C136
33pF
C137
33pF
C138
33pF
R130
8.2K
SIMIO1
R160
8.2K
SIMIN1
SIMVCC1
SIMRST
SIMCLK
TP43
TP60
SIMCLK
SIMRST
R139
0R
C77
100nF
U16
TPS2041BQDBVRQ1
IN
OUT
EN
GND
OC
C75
100nF
C76
22uF
R157
0R
TP44
LTE_VBUS
LTE_VBUS_EN
Q1
DTC043ZEBTL
Q2
DTC043ZEBTL
R37
0R
LTE_ON_OFF
R38
DNP
R39
0R
LTE_SHUTDOWN
R40
DNP
TP33
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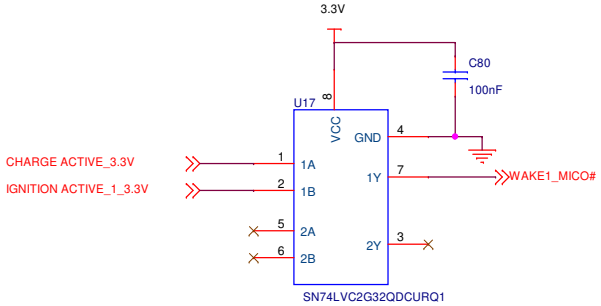




DIGITAL INPUT

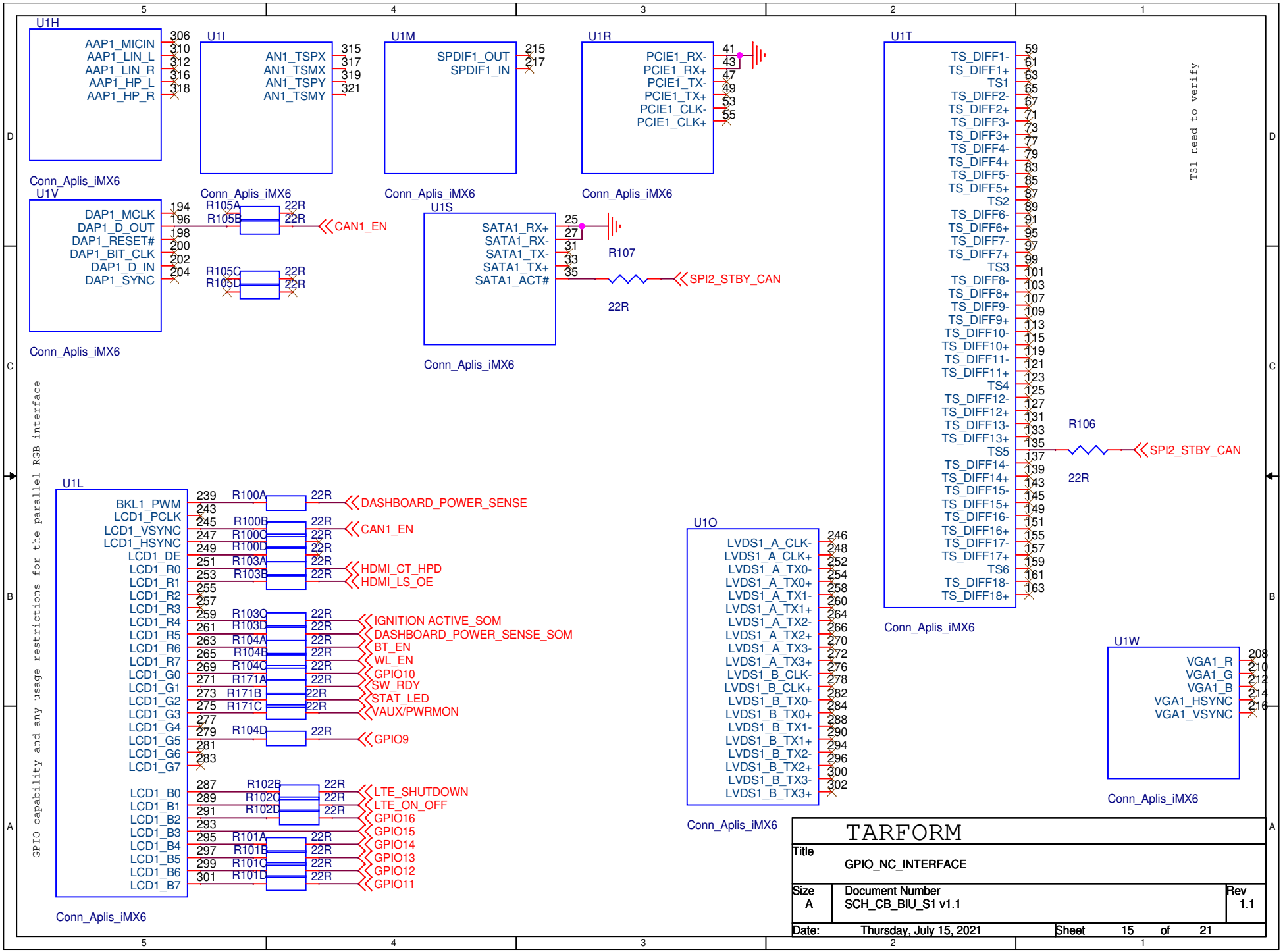


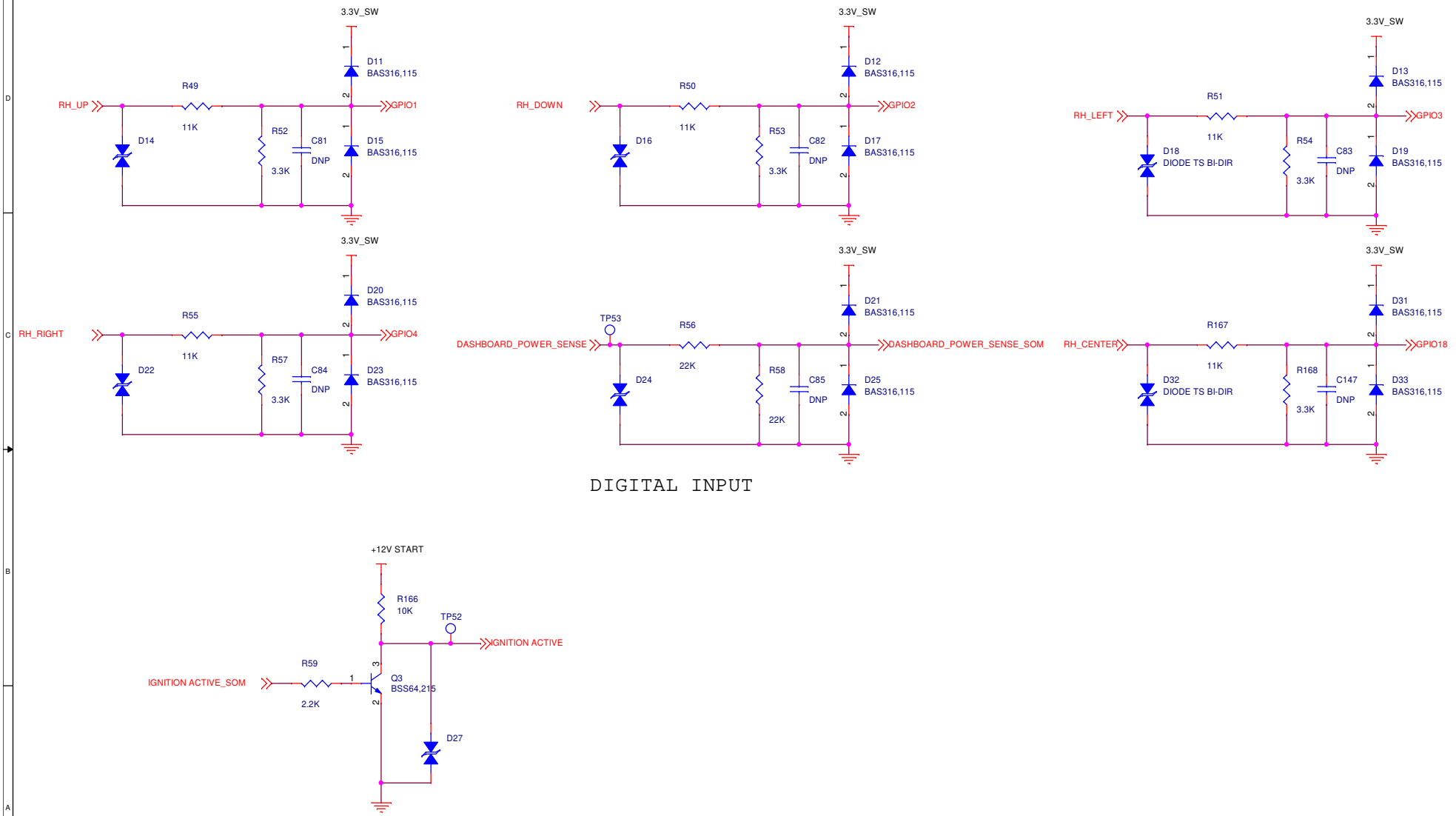
DIGITAL INPUT

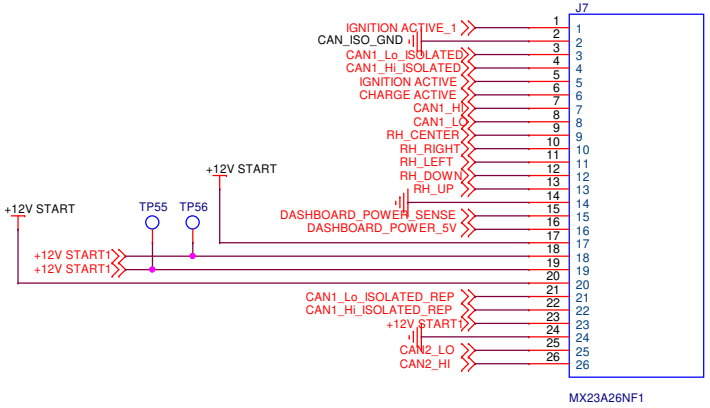


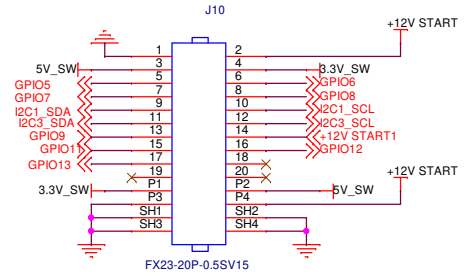
To wake up SOM module

TARFORM		
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R103D 22R
DASHBOARD_POWER_SENSE_SOM
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R104B 22R
WL_EN
R104C 22R
GPIO10
R171A 22R
SW_RDY
R171B 22R
STAT_LED
R171C 22R
VAUX/PWRMON
R104D 22R
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R102B 22R
LTE_SHUTDOWN
R102C 22R
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R102D 22R
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GPIO15
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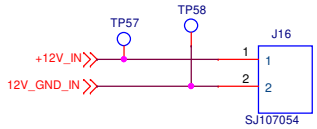

3.3V_SW
RH_UP
R49
11K
D14
R52
3.3K
C81
DNP
D11
BAS316,115
D15
BAS316,115
GPIO1
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RH_DOWN
R50
11K
D16
R53
3.3K
C82
DNP
D12
BAS316,115
D17
BAS316,115
GPIO2
3.3V_SW
RH_LEFT
R51
11K
D18
DIODE TS BI-DIR
R54
3.3K
C83
DNP
D13
BAS316,115
D19
BAS316,115
GPIO3
3.3V_SW
RH_RIGHT
R55
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D22
R57
3.3K
C84
DNP
D20
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D23
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3.3V_SW
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TP53
R56
22K
D24
R58
22K
C85
DNP
D21
BAS316,115
D25
BAS316,115
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3.3V_SW
RH_CENTER
R167
11K
D32
DIODE TS BI-DIR
R168
3.3K
C147
DNP
D31
BAS316,115
D33
BAS316,115
GPIO18
DIGITAL INPUT
+12V START
R166
10K
TP52
IGNITION ACTIVE
IGNITION ACTIVE_SOM
R59
2.2K
Q3
BSS64,215
D27
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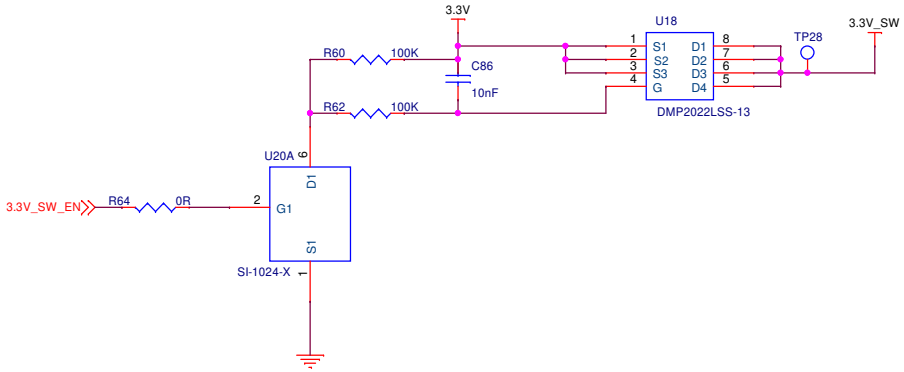


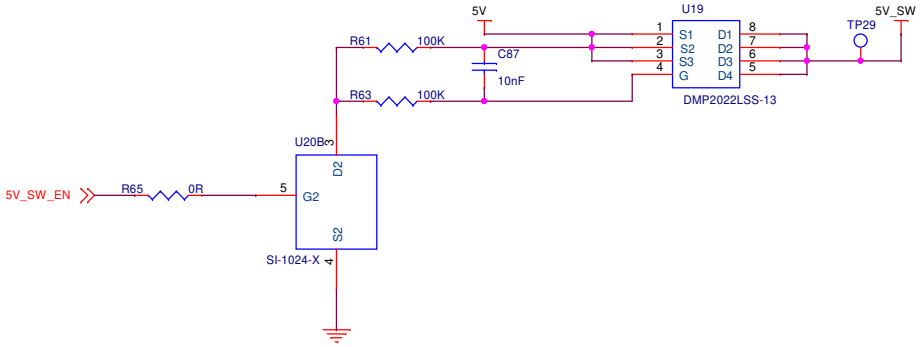
MAIN BOARD TO SECONDARY BOARD CONNECTOR

BOARD TO BOARD CONNECTOR

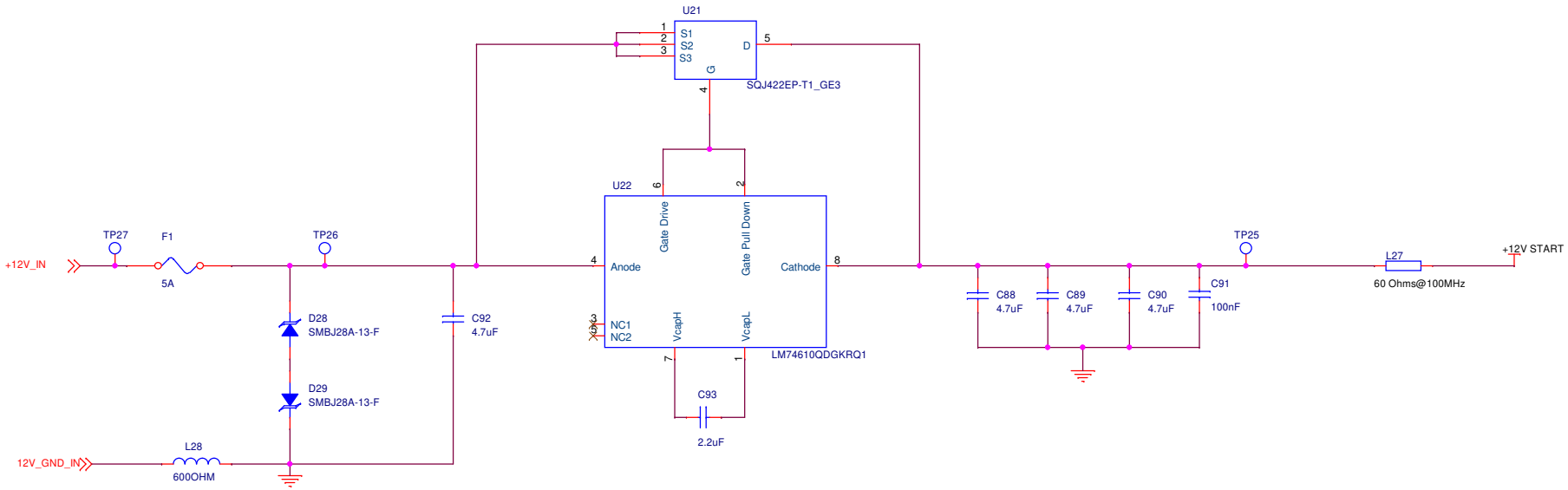






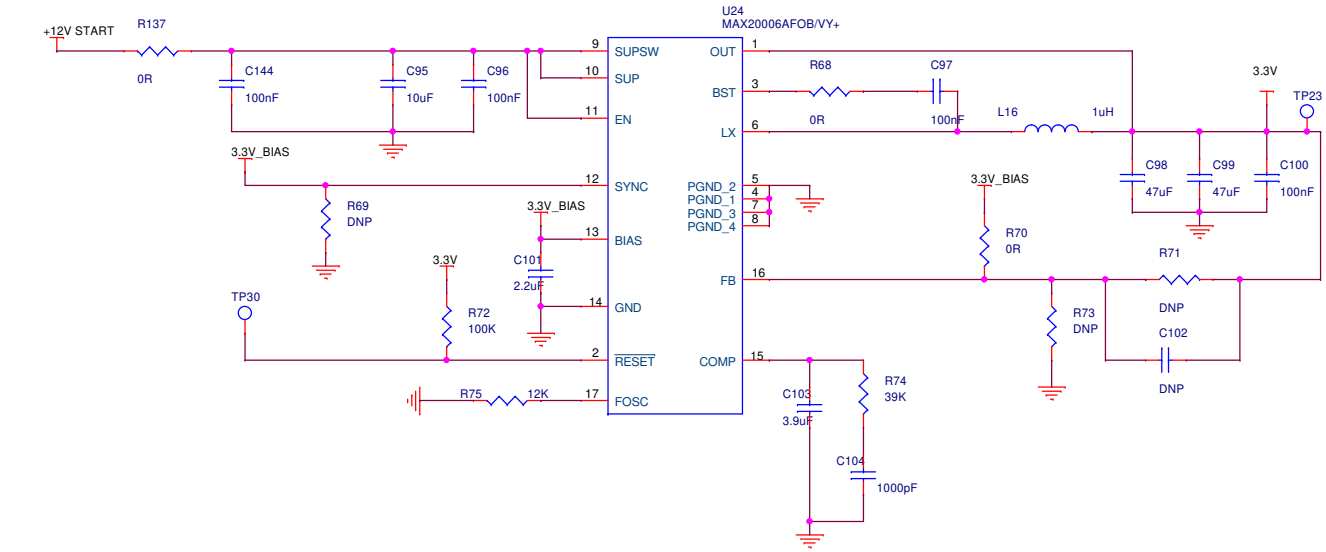


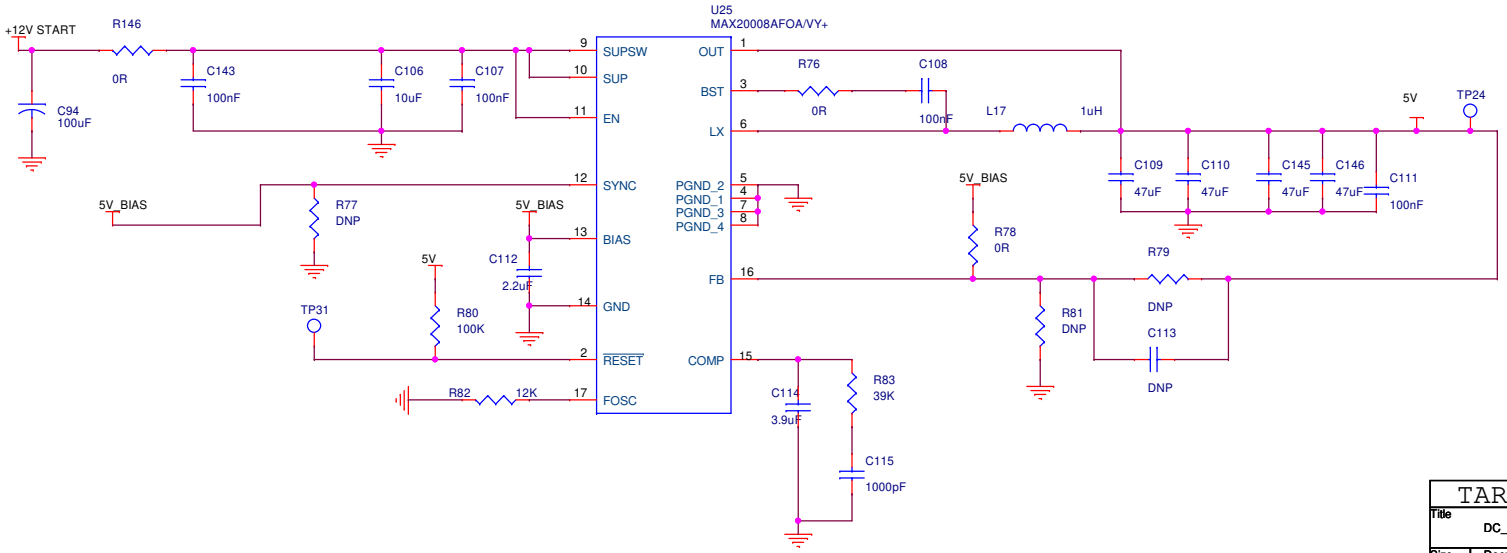
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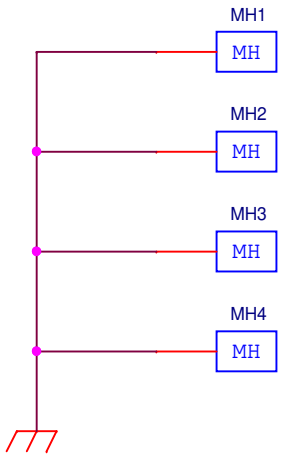


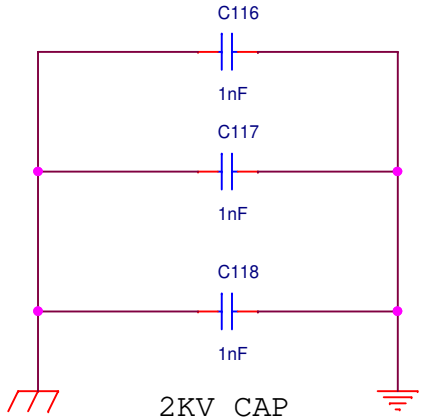


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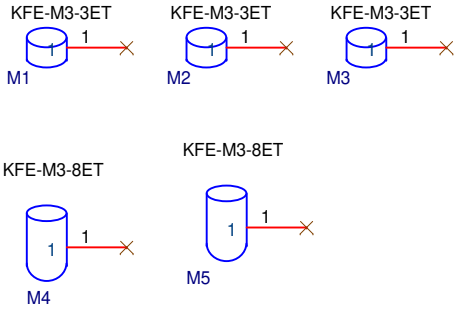
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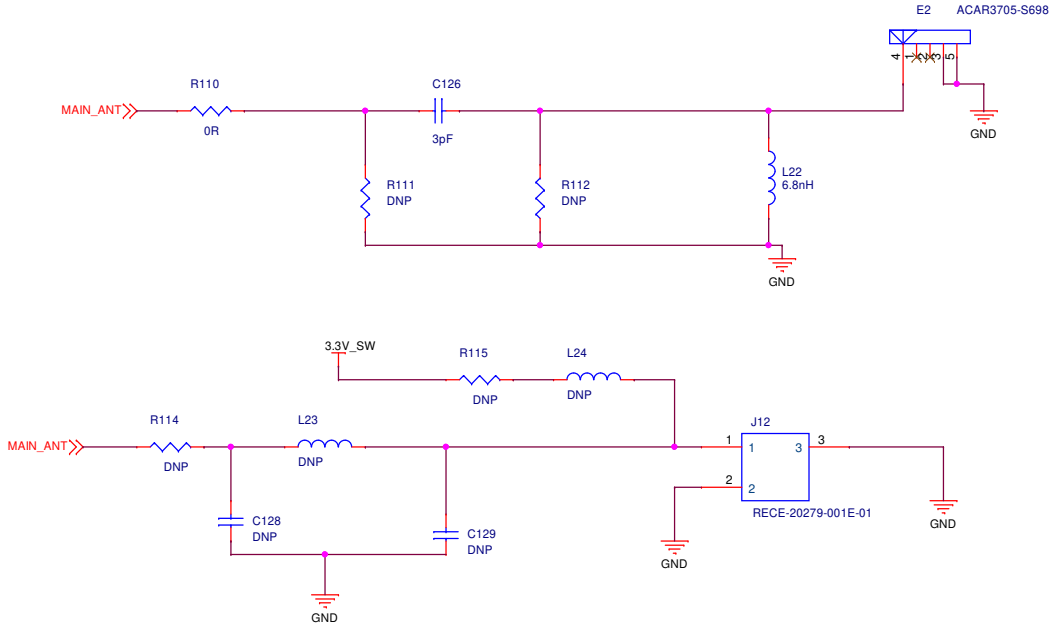




Stitching Capacitor



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